

Integration (Riemann)

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1 Motivation

The Riemann integral assigns to a bounded function $f : [a, b] \rightarrow \mathbb{R}$ a single number $\int_a^b f$ that represents the signed area between the graph of f and the x -axis. The construction is purely metric: chop $[a, b]$ into finitely many subintervals, approximate f on each subinterval by its supremum (overestimate) and infimum (underestimate), and demand that the two estimates converge as the chopping becomes finer.

2 Partitions and Darboux Sums

Definition 1 (Partition). A partition of $[a, b]$ is a finite ordered set $P = \{x_0, x_1, \dots, x_n\}$ with

$$a = x_0 < x_1 < \dots < x_n = b.$$

Set $\Delta x_i = x_i - x_{i-1}$ and define the mesh $\|P\| = \max_i \Delta x_i$. A partition P' refines P if $P \subseteq P'$.

Definition 2 (Upper and lower sums). Let $f : [a, b] \rightarrow \mathbb{R}$ be bounded. For a partition P as above, set

$$M_i = \sup_{x \in [x_{i-1}, x_i]} f(x), \quad m_i = \inf_{x \in [x_{i-1}, x_i]} f(x).$$

The upper Darboux sum and lower Darboux sum are

$$U(f, P) = \sum_{i=1}^n M_i \Delta x_i, \quad L(f, P) = \sum_{i=1}^n m_i \Delta x_i.$$

Lemma 1. If P' refines P , then $L(f, P) \leq L(f, P') \leq U(f, P') \leq U(f, P)$. Consequently $L(f, P_1) \leq U(f, P_2)$ for any two partitions P_1, P_2 .

Definition 3 (Darboux integrability). Define the upper and lower integrals

$$\overline{\int_a^b} f = \inf_P U(f, P), \quad \underline{\int_a^b} f = \sup_P L(f, P).$$

We say f is Riemann (Darboux) integrable on $[a, b]$ if the two coincide; the common value is denoted $\int_a^b f(x) dx$.

Proposition 2 (Cauchy criterion). A bounded f is integrable on $[a, b]$ iff for every $\varepsilon > 0$ there exists a partition P with $U(f, P) - L(f, P) < \varepsilon$.

Proof. ($\Rightarrow$) Suppose $\overline{\int} f = \underline{\int} f =: I$. By the definitions of inf and sup, pick P_1, P_2 with $U(f, P_1) < I + \varepsilon/2$ and $L(f, P_2) > I - \varepsilon/2$. The common refinement $P = P_1 \cup P_2$ satisfies $U(f, P) \leq U(f, P_1) < I + \varepsilon/2$ and $L(f, P) \geq L(f, P_2) > I - \varepsilon/2$, so $U(f, P) - L(f, P) < \varepsilon$.

($\Leftarrow$) For every ε we have $0 \leq \overline{\int} f - \underline{\int} f \leq U(f, P) - L(f, P) < \varepsilon$. Hence the difference is 0. $\square$

3 Continuity Implies Integrability

Theorem 3. *Every continuous function $f : [a, b] \rightarrow \mathbb{R}$ is Riemann integrable.*

Proof. Because $[a, b]$ is compact, f is bounded and uniformly continuous (Heine–Cantor). Fix $\varepsilon > 0$. By uniform continuity, choose $\delta > 0$ such that

$$x, y \in [a, b], |x - y| < \delta \implies |f(x) - f(y)| < \frac{\varepsilon}{b - a}.$$

Take any partition $P = \{x_0, \dots, x_n\}$ with mesh $\|P\| < \delta$.

On each subinterval $[x_{i-1}, x_i]$, the function f attains its sup M_i and inf m_i (continuous on a compact set), say at points $\xi_i, \eta_i \in [x_{i-1}, x_i]$. Since $|\xi_i - \eta_i| \leq \Delta x_i < \delta$,

$$M_i - m_i = f(\xi_i) - f(\eta_i) < \frac{\varepsilon}{b - a}.$$

Summing over i ,

$$U(f, P) - L(f, P) = \sum_{i=1}^n (M_i - m_i) \Delta x_i < \frac{\varepsilon}{b - a} \sum_{i=1}^n \Delta x_i = \frac{\varepsilon}{b - a} \cdot (b - a) = \varepsilon.$$

By Proposition ??, f is integrable. □

4 The Fundamental Theorem of Calculus

Theorem 4 (FTC, Part 1). *Let $f : [a, b] \rightarrow \mathbb{R}$ be continuous, and define $F : [a, b] \rightarrow \mathbb{R}$ by $F(x) = \int_a^x f(t) dt$. Then F is differentiable on (a, b) and $F'(x) = f(x)$.*

Sketch. For $h \neq 0$, $\frac{F(x+h) - F(x)}{h} = \frac{1}{h} \int_x^{x+h} f(t) dt$. Bound the integrand by $f(x) \pm \varepsilon$ using continuity of f at x , and let $h \rightarrow 0$. □

Theorem 5 (FTC, Part 2). *If f is continuous on $[a, b]$ and G is any antiderivative of f , then $\int_a^b f(t) dt = G(b) - G(a)$.*

Sketch. Both F from Part 1 and G are antiderivatives of f , so they differ by a constant: $G(x) = F(x) + C$. Then $G(b) - G(a) = F(b) - F(a) = \int_a^b f - 0$. □

5 Worked Example

For $f(x) = x^2$ on $[0, 1]$ with equal partition $x_i = i/n$,

$$U(f, P_n) = \sum_{i=1}^n \frac{i^2}{n^2} \cdot \frac{1}{n} = \frac{(n+1)(2n+1)}{6n^2} \xrightarrow{n \rightarrow \infty} \frac{1}{3}.$$

By FTC2 the antiderivative $x^3/3$ also yields $1/3 - 0 = 1/3$.